Engineering & Research Project Portfolio

Prepared by: Navya Gadhiraju

Introduction

Hello! This portfolio document compiles a comprehensive and technically rigorous record of my academic engineering projects, clinical research, and industry co-op placements. Organized in reverse chronological order from 2026 to 2024, each deep-dive section provides a technical description for each project. Projects done during a work term are organized under that term.

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Clinical Research Intern: Projects #10, #11, #12, and #13

BC Cancer, Deeley Research Centre

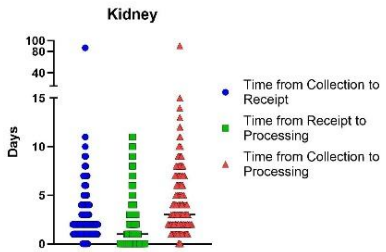
May - August 2026

At the Deeley Research Centre within the Fonseca Lab, led by Dr. Nicolette Fonseca, my work aided research on precision oncology, combining high-dimensional bioinformatics with wet-lab clinical biobanking operations.

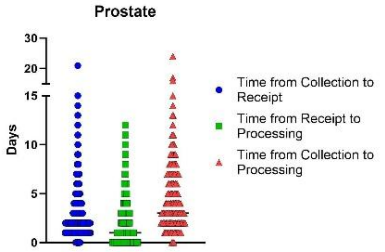
Project #10: GUBB Efficiency Representation

A critical challenge in clinical proteomics is that delays between blood collection and final laboratory processing can severely compromise the stability of circulating cell-free DNA (cfDNA) and protein biomarkers in patient samples. To systematically evaluate this provincial biobank efficiency, I analyzed hospital and LifeLabs receipt metadata across approximately 9,000 patient samples from the BC Genitourinary Cancer Biobank (GUBB) spanning a nine-year historical period from 2017 to 2026.

Moving GUBB sequencing logs into dedicated prostate and kidney sheets, I programmed in Excel to calculate the exact elapsed days for three critical operational windows: the time elapsed from collecting a patient's blood to receipt in the lab (CR), the time from receipt to final processing (RP), and the total collection-to-processing timeframe (CP). These calculations were categorized into three distinct historical segments: 2016–2019, 2020–2023, and 2024–2026. Because the total CP timeline was not the direct mathematical sum of the CR and RP windows, I represented the median times as a triple bar graph (upon a strategic design recommendation by Oscar Grainger to avoid a stacked bar graph). To visualize the statistical spread and maximum ranges of these delays, the dataset was subsequently transferred to GraphPad Prism, where final scattered box plots were plotted by my colleague Aleksandra (Ola) Przydatek. This biobank data demonstrated that a median processing delay of 3 days existed provincial-wide, establishing the clinical necessity for an in-lab experiment to assess variation in proteomic profiles over a 4-to-5-day period.



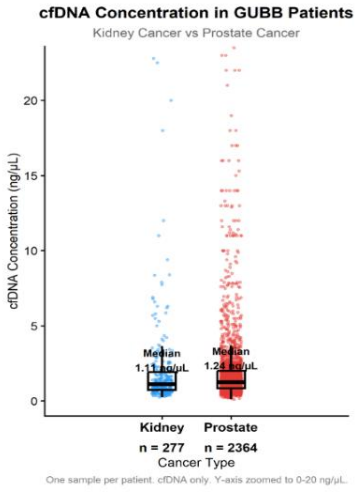
	Time from Collection to Receipt	Time from Receipt to Processing	Time from Collection to Processing
Number of values	712	712	712
Minimum	0.000	0.000	0.000
Maximum	87.00	11.00	91.00
Range	87.00	11.00	91.00
Mean	2.628	1.535	4.163
Std. Deviation	3.748	1.906	4.295
Std. Error of Mean	0.1405	0.07143	0.1610

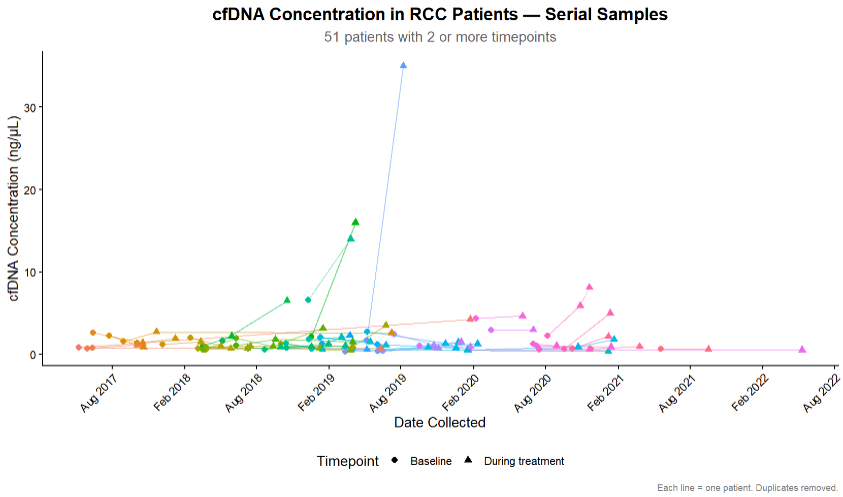


	Time from Collection to Receipt	Time from Receipt to Processing	Time from Collection to Processing
Number of values	4280	4280	4280
Minimum	0.000	0.000	0.000
Maximum	21.00	12.00	24.00
Range	21.00	12.00	24.00
Mean	2.559	1.705	4.265
Std. Deviation	2.025	1.969	2.840
Std. Error of Mean	0.03096	0.03009	0.04341

Project #11: cfDNA Concentration Median and RCC Subgroup Analysis

To further investigate biomarker characteristics, I programmed data-cleaning pipelines in R Studio (utilizing a script developed with Claude's assistance) to analyze cfDNA concentrations across GUBB patients. I structured the code to separate the "Diagnosis" column into "Prostate" and "Kidney" cohorts across multiple GUBB spreadsheets. To ensure strict data integrity, the pipeline utilized regular expression string filtering (grepl) to rename column headers, drop entries missing vital timeline parameters, and exclude patient duplicates who had multiple sample collection dates. This rigorous filtration process revealed that out of 409 renal cell carcinoma (RCC) patients, 277 represented unique individuals, while out of 4,795 prostate cancer records, 2,366 were unique. Running the R script, I plotted a scattered box plot displaying cfDNA concentrations on the y-axis and cancer type on the x-axis, which mapped a median cfDNA concentration of 1.14 ng/µL for the RCC cohort and 1.23 ng/µL for the prostate cohort. Keeping duplicate patients with different collection dates specifically for serial sample tracking, I performed an RCC subgroup analysis. Of the 277 unique RCC patients, 51 returned for multiple clinic visits over the course of their disease (43 patients with 2 visits, 5 with 3 visits, 2 with 4 visits, and 1 with 5 visits). Using R Studio, I plotted a serial sample chart mapping the fluctuations of these 51 returning patients over time.





Project #12: Blood Processing

In the wet lab, I transitioned these data insights into physical practice. To demonstrate necessary clinical precision before handling irreplaceable patient biofluids, I completed safety training and validated my micro-pipetting accuracy on an analytical scale over ten replicates. My 7 mL large-volume transfers averaged 7.036 g (Standard Deviation: 0.053), and my 460 µL small-volume transfers averaged 0.456 g (Standard Deviation: 0.005), confirming high precision. I then shadowed and assisted Ola Przydatek inside a Class II Biosafety Cabinet (BSC) during blood processing for patients with von Hippel Lindau (VHL) disease, an inherited kidney cancer.

Following SOP-001, we processed whole blood collected in Streck tubes using two-stage centrifugation. The first spin was executed in an Eppendorf 5810R or ThermoFisher Sorvall ST1 Plus centrifuge at 1600 × g for 15 minutes, separating the blood into three distinct density layers: top plasma, a thin middle white blood cell buffy coat, and bottom red blood cells. Using a Drummond Multispeed XP Portable Pipet-Aid fitted with a 5 mL serological pipette, I assisted in drawing off the plasma and isolating the gDNA-containing buffy coat. The plasma then underwent a second, higher-speed spin at 3800 × g for 10 minutes to pellet any residual cell fragments, producing a cleared plasma fraction containing cfDNA for liquid biopsies. Tube labeling was automated via a Brady Label Printer i3300.

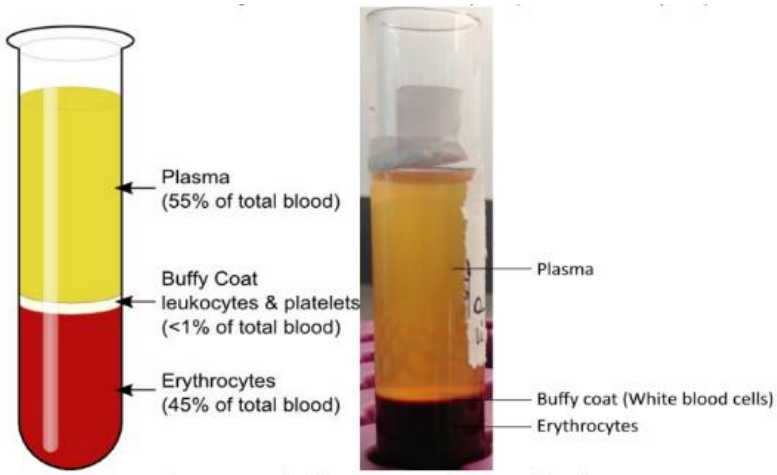


Figure 3.4.1 Blood components after centrifugation.
Reference: https://en.wikipedia.org/wiki/Blood_fractionation

Project #13: RNeasy FFPE Kir Protocol

Finally, I authored a comprehensive step-by-step visual protocol slide deck for total RNA extraction from Formalin-Fixed Paraffin-Embedded (FFPE) tumor tissue using the QIAGEN RNeasy FFPE kit. This guide mapped out deparaffinizing tissue, enzymatically digesting proteins with Proteinase K to release trapped RNA, applying heat treatment to reverse formalin-induced crosslinking, removing DNA with DNase, binding RNA to a silica spin column, and eluting pure RNA in a 2.5-hour workflow using a 5-tube labeling scheme (TISSUE, CLEAR, SUPERNATANT, FINAL RNA, BACKUP)..

- **Tube 1: `TISSUE`** — Holds your cut slices and deparaffinization solution. Used to physically melt away paraffin.
- **Tube 2: `CLEAR`** — For transferring and isolating the clear liquid phase away from the floating top wax.
- **Tube 3: `SUPERNATANT`** — Holds the clean solution after ice incubation and heavy centrifugation. Leftover debris pellet is discarded.
- **Tube 4: `FINAL RNA`** — The clean 1.5 mL collection tube from the kit box. Holds your purified sample on ice.
- **Tube 5: `BACKUP`** — An extra sterile 1.5 mL tube kept on your rack for any overflow or emergency.

I also independently went to Science Stores to procure, inventory, and store TrueBlack Lipofuscin Autofluorescence Quencher for spatial imaging assays. My work in this role developed core proficiencies across Clinical Bioinformatics (R Studio programming, database cleaning and filtering, regular expression string parsing, statistical plotting), Wet-Lab Sample Processing (clinical safety training, Class II Biosafety Cabinet operations, blood fractionation via two-stage centrifugation, plasma and buffy coat isolation, pipetting validation, laboratory logistics, reagent procurement), and Technical SOP Authoring (designing visual laboratory training protocols for QIAGEN RNeasy FFPE total RNA tissue extraction).

Research Assistant

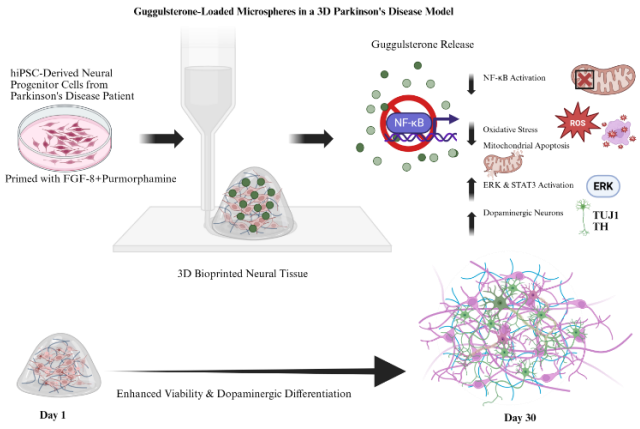
Project #9: Protocols Paper

Willerth Lab

January to April 2026

During my time as a Research Assistant in Dr. Stephanie Willerth's lab at the University of Victoria (January-April 2026), I focused on standardizing complex tissue engineering workflows to establish reproducible biofabrication standards. Working directly from a primary scientific manuscript, I co-authored a standardized laboratory protocol for 3D bioprinted neural tissue models for Parkinson's disease by adapting and translating complex experimental procedures into the formal *Sharma Current Protocols* format.

To ensure the protocol provided a clear, end-to-end execution path, I systematically documented step-by-step methodologies across all core stages of the workflow: beginning with patient-derived human induced pluripotent stem cell (hiPSC) maintenance and culture, progressing through specialized bioink formulation and microfluidic 3D bioprinting on the Aspect Biosystems RX1 platform, and concluding with downstream immunochemistry validation procedures. Throughout this process, I collaborated iteratively with team researchers to refine protocol parameters and eliminate procedural ambiguities, directly strengthening the documentation's consistency and scientific reproducibility. This completed protocol is currently under peer review for publication.



Biomedical Systems Engineer: Projects #6, #7, and #8

Victoria Hand Project

January to April 2026

Prosthetic Mechanical Design, Assembly & Repair:

To resolve recurring manufacturing and assembly bottlenecks across VHP hand systems, I led targeted mechanical troubleshooting and component updates. I identified a critical documentation error in the assembly manual where an inverted thumb piston orientation caused severe motion restriction. I corrected the CAD schematics and physical documentation, then retrofitted affected hands, including a demo unit, to restore full functional articulation. Addressing loose pin-hole tolerances on laser-cut VC300 digits 1-4 and links 3-4, I secured 6 mm pins using JB Weld epoxy as an immediate fix before systematically logging tight versus loose tolerance fits for new component batches across VC300, VO300, and VS300 systems. Following a VO300 palm redesign that rendered existing pin lengths obsolete, I updated manual specifications to a standardized 28 mm pin length. Finally, I rebuilt a damaged demo hand by replacing broken internal wrist components and evaluated PETG as a fracture-resistant alternative to standard PLA palms, targeting high-stress joint boundaries prone to snapping.



7.3 Thumb Assembly Process

First, attach the Thumb and the Thumb Piston with a 10mm pin. Ensure that the flat side of the Thumb Piston is facing upwards, and the curved edges are pointing down.

Figure 60: Attach Thumb to Thumb Piston using a 10mm pin

Attach the other side of the Thumb Piston to the Thumb Coupler on the Adaptive Grasp using a 10mm pin as shown below in Figure 61.





Project #6: Silicone Finger Mold Material Optimization

Casting silicone fingertips using rigid PLA molds created major production friction, as the rigid material made pin insertion and cured part removal difficult. To optimize this workflow, I evaluated alternative polymer molds printed in TPU and polypropylene. Through iterative casting trials, I identified polypropylene as the optimal material; its structural flexibility permitted effortless part removal and required minimal Vaseline release agent compared to PLA, while avoiding the excessive flexibility observed in TPU molds.



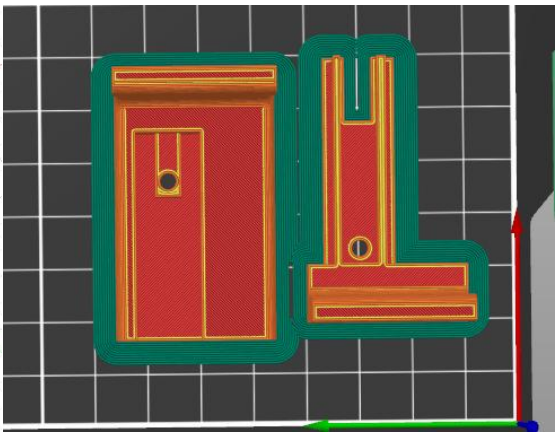
Project #7: TPU Print-Orientation Tensile Testing

To evaluate how additive manufacturing parameters influence mechanical fatigue limits, I conducted universal tensile testing on twelve Prusa thermoplastic polyurethane (TPU) coupon specimens, split evenly between horizontal and vertical print orientations. I mounted each coupon into a universal testing apparatus to gather raw force-displacement data. Following a computer data loss incident that corrupted data for two upright samples, I printed fresh replacements and completed the re-testing matrix to establish structural validation thresholds between horizontal and vertical print layer alignments.

Project #8: Mouse-Clicker Assistive Device CAD Redesign

I refined a student-designed external mouse-clicker accessibility adapter in Autodesk Fusion 360 to transform a non-functional prototype into a production-ready device. The original model suffered from overlapping sketch profiles, redundant extrusions, and non-functional screw holes caused by extruding only one of four sketch layers. I cleaned up the sketch hierarchy, resized the fastener holes to fit standard hardware, and modified the slot and top bump geometry to achieve a secure, snap-fit physical connection.





Manufacturing Operations & Onboarding Documentation:

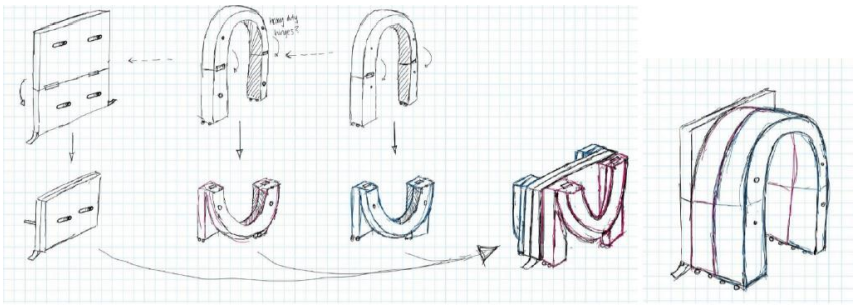
In additive farm maintenance, I repaired a Prusa XL printer by sourcing and printing a replacement back cover alongside custom nozzle alignment tools, while managing belt tension adjustments and acoustic troubleshooting across Prusa i3 MK3S+ and Ultimaker S5 systems via the Prusa web application. For workflow and file management, I validated VHP workflow software releases (V03.03.26, V03.03.27, V03.03.2026) against patient scan data, manually converting files across 3MF, bgcode, and G-code formats while scaling print models for MK4, XL, and Ultimaker hardware. Regarding documentation and logistics, I authored a comprehensive onboarding resource guide for incoming co-ops detailing assembly protocols, such as temporary pin taping, precision sanding to clear cured epoxy binding, and base-screw torque requirements during QA—while managing outreach shipment preparation, spring spray-painting, and package logistics.

Project #5: Pediatric MRI Simulator

WEST x Fraser Health Authority

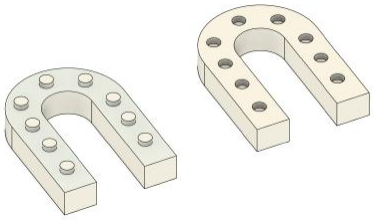
September 2025 to Spring 2026

The "Discovery Dome" was developed in collaboration with Surrey Memorial Hospital and the Fraser Health Authority to address pediatric claustrophobia and distress during loud, confined magnetic resonance imaging (MRI) procedures. Because young patients are frequently sedated before an MRI, which increases hospital recovery times and strains clinical staffing, we designed a structural, child-friendly MRI simulator to familiarize children with MRI sights and sounds. The project was bound by hospital constraints: an assembled length under 6.5 feet, a maximum width under 91.44 inches, a tunnel diameter between 60 and 70 cm, a maximum bed width of 30 inches, a collapsed size under 6 feet × 4 feet × 53 inches, a patient load capacity of at least 200 lbs, a budget of $10,000 CAD, and single-person manual assembly in under 5 minutes without risk of injury.



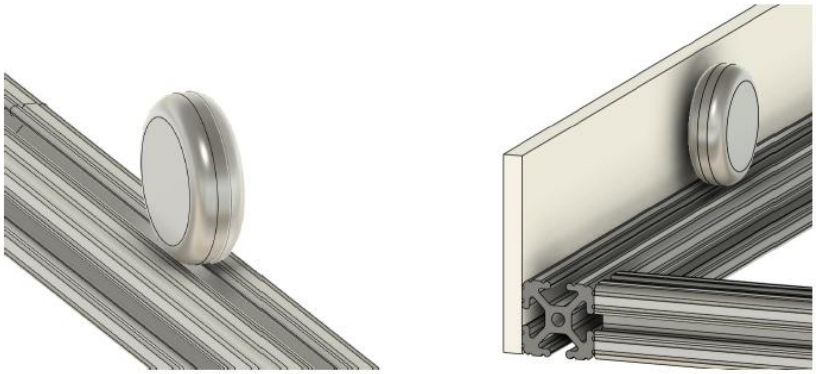
Our initial conceptual design featured a snapping Lego-style dome made of two halves that folded horizontally, with a completely open bottom section where a bed could roll in. During the design development phase, we reworked this concept to remove the open bottom, replacing it with a robust, self-contained "box" structure that provides parts storage and a stable weight-bearing foundation. We also transitioned the dome halves to fold vertically rather than horizontally, allowing for compact, single-person transport and storage without requiring a heavy, separate backing board.

For materials, the dome halves were designed an conceptualized with Glass Fiber Reinforced Plastic (GRP) made from E-glass fiber and a thermoset vinyl ester resin, finished with a chemical-resistant gelcoat to withstand sterilization and hospital-grade chemical disinfectants. To eliminate light and sound leaks, we engineered a high-precision tongue-and-groove joint (mating extruded circles on one dome edge into hollow grooves on the other) sealed with a continuous rubber gasket. The GRP dome halves securely latch onto the uPVC panels and aluminum T-slot frame of the bottom box using toolbox-style draw latches equipped with pinch-free rubber coverings.

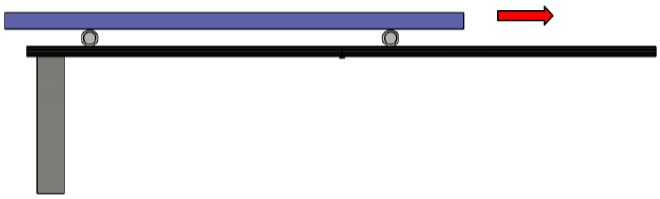


The sliding bed system underwent a major mechanical evolution. The original bed design sat on the box and utilized a singular folding leg, which required aligning two separate rail systems on the box and bed during setup. To prevent rail misalignment and sliding jams, we engineered a single, continuous bed frame that folds in half using heavy-duty midpoint hinges. The bed frame slides smoothly on V-slot aluminum extrusions using roller wheels equipped with side guard rails to eliminate pinch points and restrict horizontal play. The foot end of the table is supported by a singular, solid aluminum plate leg, which attaches to the frame using commercial aluminum folding brackets rated for over 200 lbs. These brackets contain independent locking mechanisms and soft-close speed limiters to prevent accidental collapse.

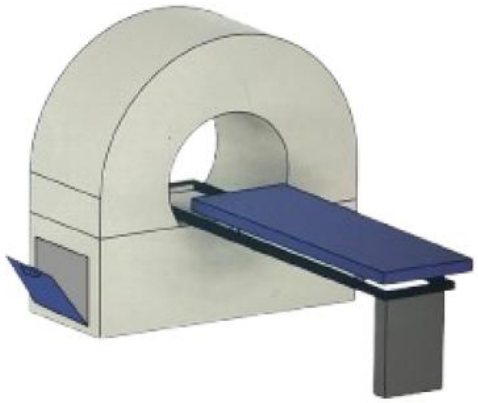
To facilitate rapid, single-person transport, we integrated a manual, retractable wheel system within a cutout cavity in the lower section of the folding bed leg. Each wheel attaches to a short vertical arm that slides along a guided track on the inner side of the leg cutout and locks into place with a manual pin, enabling the operator to roll the collapsed unit and retract the wheels to maintain a stable, non-slip base when stationary.



To ensure pediatric patient safety and comfort, the bed mat features a 2-to-3-inch thick high-density polyurethane foam core covered in a medical-grade, non-porous vinyl fabric, which is mounted onto a rigid base plate using plastic velcro strips. We designed 6-inch wide non-porous, liquid-resistant bed straps that connect to the sides of the bed with plastic snap buttons (three snap buttons per velcro strip) to allow for complete removal and chemical sanitization. The straps contain lengthwise plastic velcro strips that face inferiorly and superiorly to wrap securely around the child. For electronic servicing, we incorporated a front-access hinged door flap on the bottom box, allowing technicians to quickly access, clean, and maintain the internal vibration, auditory playback, and speaker electronics without having to disassemble the main simulator structure.



Through this project, I executed engineering tasks involving Mechanical & Mechanism Design (midpoint frame hinges, sliding tracks on V-slot aluminum extrusions, retractable wheel linkages, soft-close folding bracket selection), Materials Science for Healthcare (wet hand lay-up of E-glass/vinyl ester GRP composites, panelling, medical-grade vinyl, hospital-grade sanitization-compliant materials), and Ergonomics & Kinematics (structural loading design for 200 lbs capacity, rapid single-person assembly modeling, compact collapse geometry).



Project #4: Kelowna Biogas Plant Upgrade

WEST x FortisBC

January to April 2025

This project addressed the capacity upgrade and flow stabilization of FortisBC's Kelowna Biogas Plant (KBGP), a waste-to-energy upgrader facility located at the Glenmore Landfill. The KBGP produces renewable energy by processing captured landfill gas (LFG) and was running close to its original design limit of 500 standard cubic meters per hour (scmh). With recent increases in LFG capture, the plant required a capacity upgrade to process 800 scmh. To handle this expansion, the plant's process was reconfigured to transition from a recycling loop to a single-pass mode, feeding the gas stream directly to Pressure Swing Adsorption (PSA) units that separate methane from carbon dioxide and nitrogen. The primary mechanical engineering challenge was managing the highly variable, intermittent gas flow resulting from the cyclical regeneration of the PSA units, without over-pressurizing the existing recycle and buffer tanks or causing compressor surge.

To size the new compressor system, we conducted thermodynamic mass flow and power calculations based on plant operational data. The system required managing a peak flow rate of 421 m³/h (average flow of 278 m³/h) and achieving a goal outlet pressure of 130 psi +/- 10% (896 kPa) from an atmospheric inlet pressure of 101 kPa. The mechanical power required for this compression is determined by the pressure differential (ΔP) and the volumetric flow rate:

$$\Delta P = 896 \text{ kPa} - 101 \text{ kPa} = 795 \text{ kPa } \left(\text{kN/m}^2\right)$$

$$\text{Power} = \Delta P \times \text{Flow Rate} = 795 \text{ kN/m}^2 \times \frac{421 \text{ m}^3}{3600 \text{ s}} = 93.0 \text{ kW}$$

Factoring in a conservative mechanical efficiency of 85%, the required shaft power is:

$$\text{Horsepower} = \frac{93.0 \text{ kW}}{0.85} = 109.4 \text{ HP}$$

Compressing the LFG in a single stage without cooling would result in extreme peak temperatures. Under ideal gas laws ($P_1/T_1 = P_2/T_2$):

$$T_2 = T_1 \times \frac{P_2}{P_1} = 300 \text{ K} \times \frac{896 \text{ kPa}}{101 \text{ kPa}} = 2661 \text{ K}$$

This temperature calculation highlighted the critical safety and thermodynamic necessity of multi-stage cooling or dedicated intercoolers.

We evaluated two potential compressor solutions: an unnamed 90 kW compressor with a high flow capacity of 1,680 m^3/h (which was oversized by 400%, creating severe operational inefficiencies) and the Atlas Copco GG Variable Speed Drive (VSD) Compressor (90–132 kW power range, 75–900 m^3/h flow rate capacity, 232 psig max discharge pressure, and an operating weight of 1800–2500 kg). We recommended the Atlas Copco GG VSD because VSD technology allows the compressor to adjust its rotational speed dynamically in real-time, matching the fluctuating gas flow of the PSA regeneration cycles. This eliminated the extreme mechanical wear associated with start/stop control and avoided the severe energy losses of suction throttling, which cannot reduce gas output to zero without risking compressor damage. To optimize efficiency and protect against surge, our proposed design incorporated an anti-surge valve loop to redirect discharge gas back to the intake when mass flow fell below critical limits. It also featured high-performance alloys for wear resistance, interstage cooling water temperature reduction (which can save up to 60% of compression work), and real-time software integration using the Atlas Copco SmartLink monitoring system (tracking Service, Uptime, Energy, and AIRnet parameters).



This industrial project strengthened capabilities in Industrial Thermodynamics (mass flow rate conversions, single-stage versus multi-stage compression heat modeling, power calculations), Process Systems Sizing (sizing rotating machinery based on flow, pressure, and efficiency constraints), System Optimization (comparing Variable Speed Drives against suction throttling and start/stop controls for fluctuating chemical processes), and Industrial Safety & Control.

We later presented our project to industry leaders and faculty:



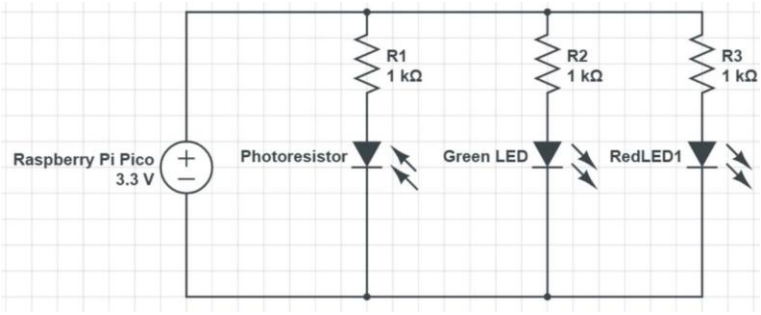
Project #3: MedaSort (Low-Cost Smart Medication Tracking System)

UVic Design and Communications II Lab:

January to April 2025

The MedaSort system was designed during my Design and Communication II Lab to mitigate clinical medication errors, unauthorized access, and temperature excursions in healthcare facilities. When temperature-sensitive medications are stored incorrectly or vials are misplaced, patient health is put at risk. Our goal was to design and manufacture a low-cost, embedded hardware-software tracking device capable of monitoring cabinet environments and logging access events in real-time.

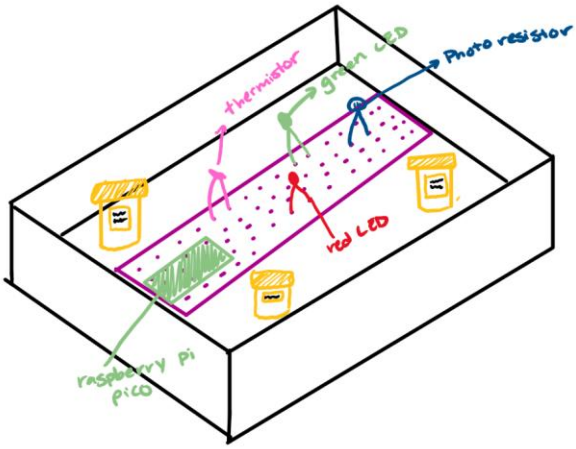
To build this tracking system, my teammate We utilized a Raspberry Pi Pico microcontroller programmed in Python. We designed, analyzed, and soldered a custom physical circuit board integrating current-limiting resistors and switching transistors to handle the signals. The system utilized two primary analog sensors: an analog thermistor to continuously log internal cabinet temperatures, and an analog phototransistor to detect changes in ambient light inside the cabinet (registering when the lid was opened or a vial was removed).



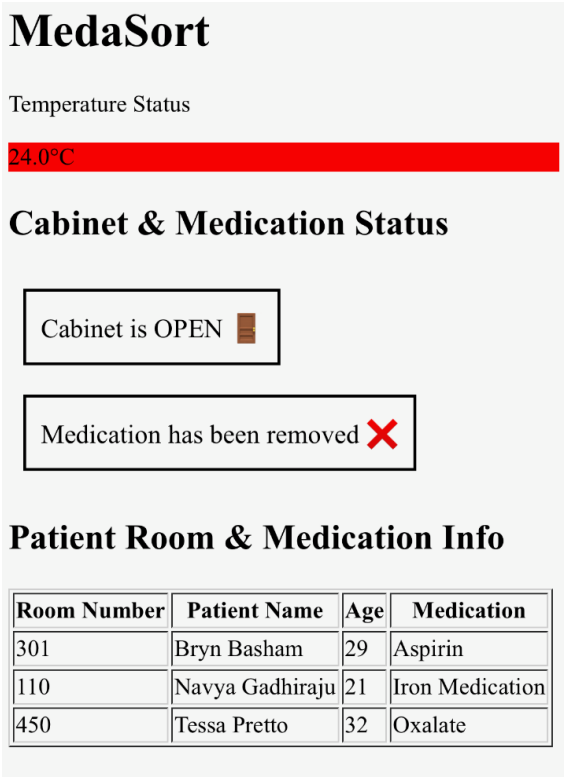
We programmed real-time Python scripts on the Pico to run two localized safety feedback loops. First, if the thermistor registered a temperature outside the safe storage threshold, the script immediately activated a local RED LED to provide an instant visual warning to clinical staff. Second, when the phototransistor detected that the cabinet was accessed, the script logged the removal event and activated a local GREEN LED to confirm a successful transaction.

Relevant components featured in MedaSort alongside their uses:

Component	Use
LED #1	Lights up green when the medication has been taken out of the box.
LED #2	Lights up red when the temperature has dropped or increased an unhealthy amount.
Thermistor	Ensures the temperature of the environment the medication is in is kept at a safe level.
Phototransistor	Detects when the medication has been removed from the compartment.



To complement the local physical indicators, we connected the hardware to an online web-based dashboard interface. This dashboard displayed patient room numbers, patient medical profiles, live room temperature readings, specific medication inventory data, and cabinet open/closed status indicators. By combining local, hardware-level visual feedback (LEDs) with network-level logging, we developed a double-layer clinical fail-safe. If the network disconnected, the local alarms remained operational; conversely, if the physical cabinet was bypassed, the automated network log captured the discrepancy, ensuring secure audit trails for sensitive clinical environments.



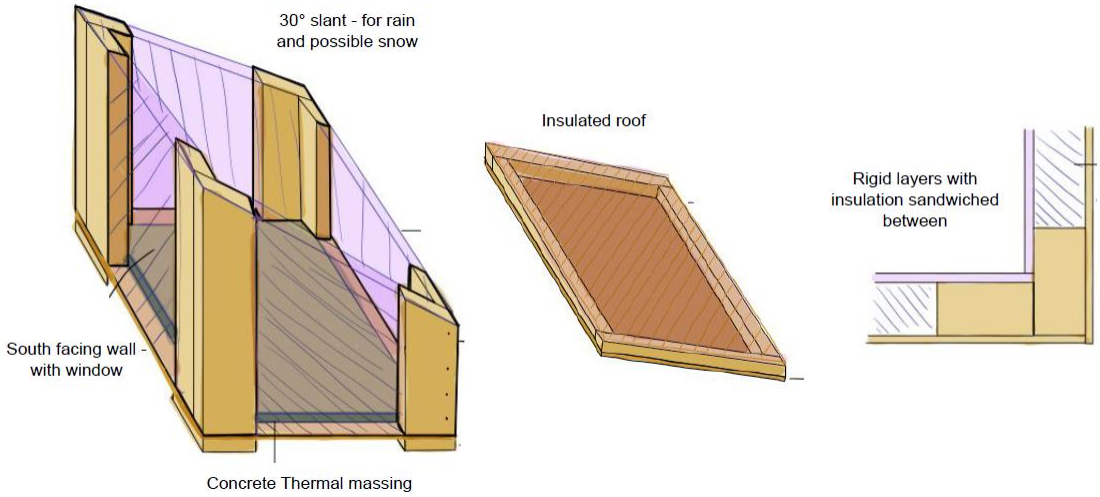
The project applied technical competencies across Embedded Systems Programming (microcontroller configuration for the Raspberry Pi Pico, real-time script development in Python), Analog Circuit Design (soldering, circuit analysis, integrating current-limiting resistors, utilizing switching transistors), Sensor Integration (calibration of analog thermistors and phototransistors), and Human-Computer Interface (HCI) design (connecting physical hardware events to a live web-based monitoring interface for healthcare environments).

Project #2: Passive Building Prototype

WEST x Introba Canada

September to December 2024

During my first semester at UVic, my team and I participated in the Women in Engineering, Science, and Technology (WEST) Passive Building Project in partnership with Introba Canada. The challenge was to design and construct a small-scale, climate-responsive passive building prototype capable of maintaining a stable internal temperature (~20°C) with no active power or electrical heating. The design had to fit within strict parameters: an exact interior volume of 0.22 m^3, a width under 30 inches (0.762 m) to easily fit through standard doors, and the integration of a donated triple-pane glass window from Cascadia Windows & Doors (rated with a U-factor of 0.14, a Solar Heat Gain Coefficient [SHGC] of 0.20, visible transmittance of 0.45, and air leakage < 0.3). The total materials budget was capped at $250 CAD.



My teammates Rana El-Shaarawi, Zoe Dancer, Sarah Shariff, and I began by researching passive building principles. We selected a prototype width of 0.425 m to easily clear the door clearance limit. We chose a shortest back height of 0.6 m and a 30° roof slope for rain runoff. To calculate the front height (H_{front}):

$$H_{\text{front}} = 0.6\ \text{m} + 0.425\ \text{m} \times \tan(30°) = 0.846\ \text{m}$$

Using 2x4 framing lumber (0.04 m thick), we calculated the inner length (l) required to meet the 0.22 m³ volume constraint:

$$(l - 0.08) \times (0.425 - 2 \times 0.04) \times 0.6 + 0.5 \times (l - 0.08) \times (0.425 - 2 \times 0.04) \times 0.245 = 0.22 \text{ m}^3$$

Solving this equation yielded $l = 0.95$ m. To accommodate 10 cm of precast concrete paver thermal mass on the floor, we selected a final length of 1.0 m. Finally, we calculated the roof width:

$$\text{Roof Width} = \frac{0.425}{\cos(30°)} = 0.490 \text{ m}$$

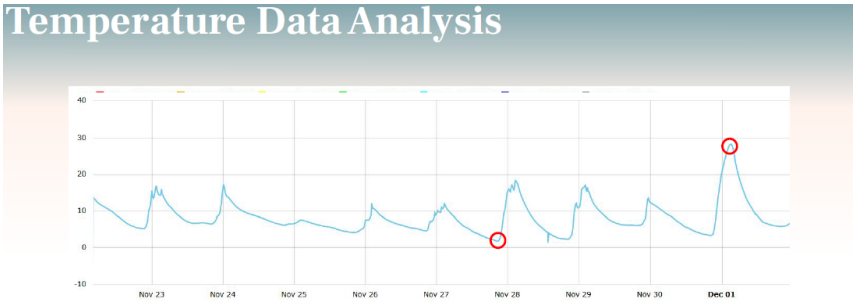
We rounded this to 0.5 m to ensure a small overhang and prevent gaps. The outside surface areas were calculated as: Floor Area = 0.3174 m², Side Area = 2.26 m², and Roof Area = 0.490 m², resulting in a total Surface Area of 3.0674 m² and an exact form factor (SA/V) of 13.9.



We fabricated the prototype with hand saws, a staple gun, and hand drills during build sessions. The total material cost was $179.97 (Plywood $82.11, hardboard $20.73, markerboard waterproof layer $23.10, foil insulation $13.50, precast concrete paver thermal mass $6.98, and sandpaper and blocks $11.13; the 2x4 framing lumber and Rockwool batts were provided). Rather than casting concrete in place, which would introduce excess moisture, we used the precast concrete paver floor. We stuffed foam backer rods into critical frame joints to eliminate air leaks.

Adhering to a Cradle-to-Cradle (C2C) lifecycle, we relied exclusively on mechanical fasteners (screws, staples), avoiding adhesives to ensure the prototype could be easily disassembled and its components reused. We tested the prototype for a week on Introba's downtown Victoria rooftop during a torrential November downpour. Despite a storm that tested our moisture barrier, the

structure maintained high thermal retention, logging a historical low of 1.9°C on November 28 and hitting an indoor peak of 28.4°C on December 1.



Historic Low		Historic High	
1.9°C	Thursday, Nov.28	28.4°C	Sunday, Dec. 1

The work incorporated Passive Architectural Design (passive solar gain via south-facing glazing, continuous building envelope insulation, thermal massing, thermal bridging reduction), Geometric & Sizing Calculations (volumetric, surface area, trigonometric slope calculations), Sustainable Life Cycle Analysis (Cradle-to-Cradle mechanical fastening design for disassembly), and Environmental Testing (analyzing outdoor temperature loggers to evaluate physical prototype insulation performance).





Project #1: High School Capstone: Podcast Series & "Period Poverty" Social Action Campaign

Prince of Wales Mini School

January to June 2024

During my final year at the Prince of Wales Mini School, serving as Feminism Club President, I designed and recorded a three-episode educational podcast series (averaging 7 minutes per episode) addressing Gender Stereotyping, Beauty Standards, and Reproductive Rights under the mentorship of biochemist and genetics PhD Dr. Agnieszka Kielczewska. Concurrently, my co-organizers and I launched the "Period Poverty" Social Action Campaign from April 14 to May 29, 2024, to combat menstrual stigma and support vulnerable demographics in our community.

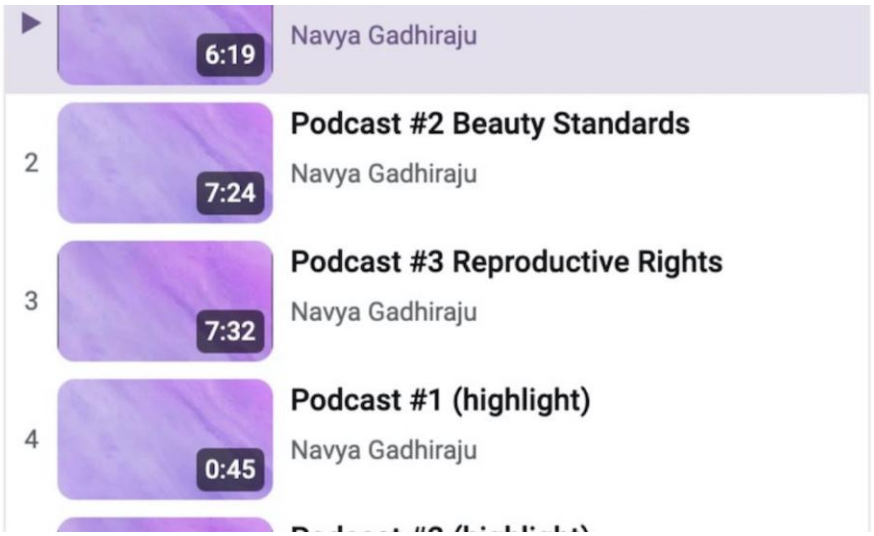
I designed informational flyers, educational posters, and parent newsletters on Canva, and established partnerships with local administrations. We delivered six interactive educational presentations and workshops to intermediate classes at Bayview Elementary (two classes) and Tennyson Elementary (four classes) to foster safe discussions about menstruation among youth. Alongside these workshops, we managed a feminine hygiene product and donation drive from May 8 to 26, collecting physical products (pads, tampons, wipes, hand sanitizer) and cash donations in designated collection boxes at the main entrances of Prince of Wales. We leveraged resources and guidance from Anjali Katta (GirlsCo) and Holly Johnston (Hello Period). At the conclusion of our drive, we donated 100% of the collected menstrual products and monetary proceeds directly to the Downtown Eastside Women’s Centre to support local women's shelters.







This capstone project refined skills in Project Management (managing educational campaigns and collection drives under tight community-partner timelines), Science & Social Communication (simplifying complex biological and social topics for intermediate elementary school classes and facilitating discussions), and Graphic Design & Outreach (creating community outreach newsletters, flyers, and posters using digital design software).



If you have any further questions for me, or would like to see references, I have the project presentations, (publication*), proposal reports, and work term reports that I would be happy to share!

*Publication is still under review